

IMO 2026 — Problem 5

Solution by Meta Muse Spark 1.1

2026-07-21

Source: `results/muse-spark-1.1/problem-05/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Let $\mathbb{R}_{>0}$ be the set of positive real numbers. Determine all functions $f : \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}$ such that $\sqrt{\frac{x^2+f(y)^2}{2}} \geq \frac{f(x)+y}{2} \geq \sqrt{xf(y)}$ for every $x, y \in \mathbb{R}_{>0}$.

Solution document

Status

solved

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Approaches tried

- Reformulated as two squared inequalities and introduced α, β with sum $2(x - f(y))^2$.
- Obtained $x = f(y)$ collapse giving $f(f(y)) = 2f(y) - y$.
- Showed $c(x) = f(x) - x \geq 0$ by iterating and positivity.
- Showed $c(f(x)) = c(x)$ and orbit is arithmetic: $f^n(y) = y + nc(y)$.
- Derived bounds for c_x around image points: $|c_x - c_y| \leq (x - f(y))^2 / f(y)$.
- Used density/separation argument to prove all positive c equal and mixed zero/positive impossible.
- Concluded $f(x) = x + c$, $c \geq 0$.

Current best

The only solutions are $f_c(x) = x + c$ for $c \geq 0$. Verification: $(f_c(x) + y)/2 = (x + y + c)/2$, $xf_c(y) = x(y+c)$, $x^2 + f_c(y)^2 = x^2 + (y+c)^2$, and both AM-GM and RMS-AM give $(x - (y+c))^2 \geq 0$.

Full proof

Let $P(x, y)$ be the statement

$$\sqrt{\frac{x^2 + f(y)^2}{2}} \geq \frac{f(x) + y}{2} \geq \sqrt{xf(y)} \quad x, y > 0. \quad (0)$$

Both sides ≥ 0 . Squaring,

$$A(x, y) : (f(x) + y)^2 \geq 4xf(y), \quad (1)$$

$$B(x, y) : 2x^2 + 2f(y)^2 \geq (f(x) + y)^2. \quad (2)$$

Indeed (0) is equivalent to (1) and (2). Set

$$\alpha(x, y) = (f(x) + y)^2 - 4xf(y) \geq 0, \quad \beta(x, y) = 2x^2 + 2f(y)^2 - (f(x) + y)^2 \geq 0.$$

Then

$$\alpha + \beta = 2x^2 + 2f(y)^2 - 4xf(y) = 2(x - f(y))^2 \geq 0. \quad (3)$$

Put $c(x) = f(x) - x$, so $f(x) = x + c(x)$ and $f(x) > 0 \Rightarrow c(x) > -x$.

1. **The image point gives equality.** Fix $y > 0$ and take $x_0 = f(y) > 0$. Then $x_0 - f(y) = 0$, so by (3) $\alpha(x_0, y) + \beta(x_0, y) = 0$. Since both ≥ 0 , $\alpha = \beta = 0$. Hence

$$(f(f(y)) + y)^2 = 4f(y)^2.$$

Both sides > 0 , so $f(f(y)) + y = 2f(y)$ and

$$f(f(y)) = 2f(y) - y \quad \forall y > 0, \quad (4)$$

in particular $2f(y) > y$, i.e. $f(y) > y/2$, and

$$c(f(y)) = f(f(y)) - f(y) = f(y) - y = c(y). \quad (5)$$

2. **Nonnegativity of c .** Iterate (4). Define $a_0 = y$, $a_{n+1} = f(a_n)$. By (5), $c(a_{n+1}) = c(a_n)$. Induction gives $c(a_n) = c(y)$ and

$$a_{n+1} = a_n + c(y), \quad a_n = y + nc(y) \quad n \geq 0. \quad (6)$$

All $a_n > 0$ because f maps to > 0 . If $c(y) < 0$ then $a_n \rightarrow -\infty$, and for $n > y/|c(y)|$ we have $a_n \leq 0$, impossible. Hence

$$c(y) \geq 0 \quad \forall y, \quad f(y) \geq y. \quad (7)$$

Moreover (5)–(6) hold.

Denote $p_y = c(y) \geq 0$, $P_y = f(y) = y + p_y$. Then $f(y) = P_y$.

3. **Local confinement of c .** From (1)(2) with $f = x + c$ notation,

$$(x + y + c(x))^2 \geq 4x(y + c(y)), \quad (x + y + c(x))^2 \leq 2x^2 + 2(y + c(y))^2. \quad (8)$$

Fix y and put $P = P_y = y + c(y)$. Then $y + c(y) = P$. (8) becomes

$$2\sqrt{xP} - x - y \leq c(x) \leq \sqrt{2x^2 + 2P^2} - x - y, \quad (9)$$

where the lower bound is effective only when it is > 0 (otherwise $c(x) \geq 0$ already). Rewrite using $y = P - c(y)$:

$$c(x) - c(y) \in [I_L(x, P), I_U(x, P)], \quad (10)$$

$$I_L(x, P) = 2\sqrt{xP} - x - P = -(\sqrt{x} - \sqrt{P})^2 = -\frac{(x - P)^2}{(\sqrt{x} + \sqrt{P})^2},$$

$$I_U(x, P) = \sqrt{2x^2 + 2P^2} - x - P = \frac{(x - P)^2}{\sqrt{2x^2 + 2P^2} + x + P}.$$

Thus $I_L \leq 0 \leq I_U$ and

$$I_L(x, P) = -(x + P - 2\sqrt{xP}), \quad |I_L| = (\sqrt{x} - \sqrt{P})^2, \quad I_U \geq 0.$$

Hence

$$|c(x) - c(y)| \leq \max(|I_L|, I_U).$$

But

$$|I_L| = \frac{(x - P)^2}{(\sqrt{x} + \sqrt{P})^2} \leq \frac{(x - P)^2}{P}, \quad I_U = \frac{(x - P)^2}{\sqrt{2x^2 + 2P^2} + x + P} \leq \frac{(x - P)^2}{P}. \quad (11)$$

Therefore for every $y > 0$ with $P = f(y)$,

$$|c(x) - c(y)| \leq \frac{(x - P)^2}{P} \quad \forall x > 0. \quad (12)$$

In particular if $x = P$, $c(x) = c(y)$, consistent with (5).

4. All positive values of c coincide. Let

$$Z = \{x : c(x) = 0\}, \quad Q = \{x : c(x) > 0\}.$$

Assume $a, b \in Q$, $p = c(a) > 0$, $q = c(b) > 0$. By (6), forward orbits

$$A_n = a + np \quad (n \geq 0), \quad B_m = b + mq \quad (m \geq 0)$$

satisfy $c(A_n) = p$, $c(B_m) = q$, and for $n \geq 1$, $A_n = f(A_{n-1}) \in \text{Im } f$, similarly $B_m \in \text{Im } f$ for $m \geq 1$.

Fix $n \geq 1$, $P_n = A_n$. For any $X > 0$ there is m with $|X - B_m| \leq q/2$ (nearest point of the arithmetic progression, at least for $X \geq b$). Choose $X = P_n$ with $P_n \geq b$ (eventually true since $P_n \rightarrow \infty$). Pick m_n minimizing $|P_n - B_{m_n}|$; then $h_n := B_{m_n} - P_n$, $|h_n| \leq q/2$.

Apply (12) with $y = A_{n-1}$ so $P = f(y) = P_n$ and $x = B_{m_n}$:

$$|c(B_{m_n}) - c(y)| = |q - p| \leq \frac{h_n^2}{P_n} \leq \frac{(q/2)^2}{P_n}.$$

Since $P_n = a + np \rightarrow \infty$, the right side $\rightarrow 0$. Hence $p = q$. So any two points with positive c have same value. Denote this common value by $p > 0$ if $Q \neq \emptyset$.

5. **Zero and positive cannot coexist.** Suppose $Z \neq \emptyset$ and $Q \neq \emptyset$, so $p > 0$ is the unique positive value. Take $b \in Z$, $a \in Q$. For pair $(x = a, y = b)$, condition B gives

$$(a + b + p)^2 \leq 2a^2 + 2b^2 \iff (a - b)^2 \geq p^2 + 2p(a + b) \geq p^2. \quad (13)$$

Thus $|a - b| \geq p$ for every $a \in Q, b \in Z$. Indeed (13) is exactly the B -inequality. Consequently

$$\inf\{|a - b| : a \in Q, b \in Z\} \geq p > 0. \quad (14)$$

If $\inf$ is $\geq d > 0$, then Z and Q are both open in $(0, \infty)$: If $z \in Z$ and $|x - z| < d$, x cannot belong to Q (otherwise distance $< d$), so the d -ball around z stays in Z ; similarly for Q . Thus $(0, \infty) = Z \sqcup Q$ would be a separation into two disjoint non-empty open sets, contradicting connectedness of $(0, \infty)$. Detailed: Suppose $d = \text{dist}(Z, Q) > 0$. For $z \in Z$, $(z - d/2, z + d/2) \cap (0, \infty)$ meets Q would give points at distance $< d$; hence the interval lies in Z . So Z open, similarly Q open.

Therefore one of Z, Q is empty.

If $Q = \emptyset$, then $c \equiv 0$.

If $Z = \emptyset$, then $c(x) = p > 0$ for all x , constant.

6. **Form of f .** In either case c is constant: $c(x) = c_0 \geq 0$ for all x . Hence

$$f(x) = x + c_0, \quad c_0 \geq 0.$$

Positivity $f : (0, \infty) \rightarrow (0, \infty)$ exactly requires $c_0 \geq 0$ (if $c_0 < 0$, choose $x < |c_0|$ giving $f(x) \leq 0$).

7. **Verification.** Let $f_c(x) = x + c$, $c \geq 0$. Then for any $x, y > 0$,

$$\frac{f_c(x) + y}{2} = \frac{x + y + c}{2},$$

$$\sqrt{xf_c(y)} = \sqrt{x(y + c)}, \quad \sqrt{\frac{x^2 + f_c(y)^2}{2}} = \sqrt{\frac{x^2 + (y + c)^2}{2}}.$$

By AM $\geq$ GM, $\frac{x + (y + c)}{2} \geq \sqrt{x(y + c)}$, i.e.

$$(x + y + c)^2 - 4x(y + c) = (x - y - c)^2 \geq 0.$$

By RMS $\geq$ AM, $\sqrt{\frac{x^2 + (y + c)^2}{2}} \geq \frac{x + y + c}{2}$, i.e.

$$2x^2 + 2(y + c)^2 - (x + y + c)^2 = (x - y - c)^2 \geq 0.$$

Thus both inequalities hold, with equalities exactly when $x = y + c = f_c(y)$.

Hence all functions $f(x) = x + c$, $c \geq 0$, and only they, satisfy the requirement.